🎧 Cluster Analysis Report – Amazon Music Clustering

📊 Overview

Using K-Means clustering on audio features, the dataset was segmented into **3 distinct clusters**. Dimensionality reduction techniques (**PCA and t-SNE**) confirm that these clusters are well-separated and meaningful.

🔵 Cluster 0 – Instrumental / Calm / Acoustic

🔍 Characteristics:

- Low energy (~0.31) → calm and relaxed tracks
- High acoustic Ness (~0.75) → natural / unplugged sound
- Low loudness (~ -13 dB) → soft audio
- Low speechiness (~0.06) → minimal vocals
- Moderate valence (~0.41) → slightly positive mood
- Tempo ~112 BPM → medium pace

🎵 Interpretation:

This cluster represents **instrumental, classical, jazz, and acoustic music**.

👤 Top Artists:

- Charlie Chaplin
- Louis Armstrong
- Bill Evans
- Charles Aznavour

🧠 Label:

Calm / Instrumental / Acoustic

🟠 **Cluster 1 – Energetic / Mainstream / Pop**

🔍 **Characteristics:**

- High energy (~0.70) → upbeat tracks
- High valence (~0.67) → happy and positive
- Moderate loudness (~ -7.6 dB) → louder songs
- Low acoustic Ness (~0.26) → electronic production
- Moderate danceability (~0.63)
- Highest tempo (~125 BPM)

🎵 **Interpretation:**

This cluster represents **mainstream, commercial, and energetic music**.

👤 **Top Artists:**

- ARASH
- Wings
- Karibe con K

🧠 Label: **Energetic / Mainstream / Pop-like**

🟢 **Cluster 2 – Speech / Vocal Heavy / Audiobook**

🔍 **Characteristics:**

- Very high speechiness (~0.84) → spoken content
- High danceability (~0.67)
- Moderate energy (~0.46)
- Higher liveness (~0.43) → live/recorded feel
- Low loudness (~ -13.48 dB)
- Lowest tempo (~100 BPM)

🎵 **Interpretation:**

This cluster represents **audiobooks, storytelling, and speech-heavy content**.

👤 **Top Artists:**

- Die Drei???
- Bibi Blocks Berg
- Benjamin Blüm Chen

🧠 Label: **Speech / Audiobook / Vocal-heavy**

📌 Key Insights

- **Speechiness** strongly separates audiobook content (Cluster 2)
- **Energy and tempo** define energetic/mainstream music (Cluster 1)
- **Acoustic Ness** identifies calm/instrumental tracks (Cluster 0)

📈 Model Validation

- **PCA Visualization** → clear linear separation
- **t-SNE Visualization** → strong non-linear separation
- **Cluster Heatmap** → distinct feature patterns across clusters

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🎯 Conclusion

The clustering successfully identified **three meaningful listener/music segments**:

1. Calm / Instrumental listeners
2. Energetic / Mainstream listeners
3. Speech / Audiobook listeners

This segmentation can be directly used for:

- 🎧 Music recommendation systems
- 👤 User personalization
- 📊 Content categorization